

PYS-01 • BIOMEDICAL

Magnetic Capsule Maze Cup

*Steer a magnetic pill-camera through a body
maze without touching it.*

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN

Fastest clean run with low penalties and a clear magnetic-control
explanation.

PYS-02 • MATERIALS

Oobleck Armor Arena

*Design armor with a material that acts like a
liquid until impact.*

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Best target protection with least material and cleanest reset.

PYS-03 • MECHANICS

Cardboard Automata Arcade

*Build a one-crank machine that performs a mini
task.*

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Most reliable motion with useful mechanism and clear
explanation.

PYS-04 • BIOMEDICAL

Stethoscope Sprint and Recovery

*Build a medical listening tool and use it like a
sports scientist.*

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Best sound quality plus accurate, respectful data collection.

PYS-05 • BIOMEDICAL

Reaction Time Combine

*Train like a sports scientist and prove which
strategy cuts delay.*

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Best median plus most improvement using clean data.

PYS-06 • WAVES + SOUND

SONAR Slinky Showdown

Make invisible distance sensing visible with a slinky pulse.

PREDICT › SEND PULSE › TIME N TRIPS › CALCULATE › DEFEND

WIN

Most correct wave predictions and station solutions.

PYS-07 • OPTICS + LIGHT

Pinhole Precision Challenge

Build the sharpest camera image with no lens.

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Best image clarity and concept explanation after one aperture redesign.

PYS-08 • MECHANICS

Low-Ropes Force Map Relay

*Predict balance and stability before the balance
challenge proves you right.*

PREDICT › TEST FIXED CARD › MAP FORCES › COMPARE › SCORE

WIN

Most accurate predictions and clear debrief connections.

PYS-09 • MECHANICS

Hovercraft Hockey Hackathon

Build a puck that moves by barely touching the floor.

PREDICT > BUILD or SOLVE > TEST > SCORE > REDESIGN

WIN

Highest normalized glide and five-shot target scores, with explanation and build quality.

PYS-10 • OPTICS + LIGHT

Spectra Sleuth Showdown

*Can two white lights hide different color
fingerprints?*

PREDICT > OBSERVE > SKETCH > MATCH > JUSTIFY

WIN

Most correct spectrum matches and strongest safe fireworks-
science explanation.

PYS-11 • COMPUTING

BookBot Bin Logic Challenge

*Retrieve the right book by using bin addresses, not
a conveyor race.*

PLAN › RUN SHARED MAT › COUNT MOVES › CHECK OPTIMUM ›
DEFEND

WIN

Most correct retrievals with the fewest legal moves, card-normalized efficiency, and clear algorithm defense.

PYS-12 • DESIGN

Accessibility Ramp Rescue Lab

*Design a portable ramp that works for a real user
constraint.*

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Best safe ramp meeting slope, load, portability, and client criteria.

PYB-01 • MECHANICS

Pencil Pressure Safety Lab

Spread a force before it dents the target.

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Largest area protected with clear data.

PYB-02 • BIOMEDICAL

Domino Neuron Relay

Make a nerve signal survive a gap.

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Fastest reliable signal with best explanation.

PYB-03 • MECHANICS

Pulley Rescue Relay

Lift the load with smarter force, not stronger arms.

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Best lift with correct setup and explanation.

PYB-04 • COMPUTING

Barcode Checksum Rescue

*Catch the fake barcode before the warehouse ships
the wrong part.*

LEARN RULE › CALCULATE › VERIFY › AUDIT › EXPLAIN

WIN

Most bad codes detected with few false alarms.

PYB-05 • COMPUTING

Code Break Cipher Relay

*Crack ten ciphers and race the answer to the judge
before time runs out.*

PREDICT › BUILD or SOLVE › TEST › SCORE › REDESIGN

WIN

Most points across ten codes; a perfect sweep is exactly 100.